CSWeb 8.1 on cPanel Shared Hosting

Installation and Operation Guide

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About this package

This package installs **CSWeb 8.1** (CSWeb Community edition) on a shared hosting account managed with **cPanel**. It is ready to upload: no SSH, Composer or command-line tools are required to install it.

On shared hosting, CSWeb is used for **data collection and synchronization**:

- deploying CSPro applications to tablets and phones,
- synchronizing cases between CSEntry and the server,
- monitoring progress (sync report, map),
- managing interviewers and supervisors,
- downloading the data for processing with CSPro (Data Viewer, export to Stata, SPSS, R or CSV).

The **breakout** feature (automatic export of cases to relational tables) and scheduled jobs are not available on shared hosting. They require a dedicated server.

Tested configuration

The package was installed and tested in an environment reproducing a cPanel shared hosting plan:

Component	Version / setting
PHP	8.3, with `exec`, `proc_open`, `shell_exec`, `system`, `popen` disabled
Web server	Apache 2.4 with `mod_rewrite` and `.htaccess`
Database	**MariaDB 10.6** (also compatible with MySQL 8)
Limits	1 CPU core, 20 database connections per user

Results: installation, application deployment, CSEntry synchronization (including **5,000 cases in a single sync**), data download, user management and Arabic text all work. **200 devices synchronizing within the same minute completed with no error.**

Before you start

Hosting checklist

Requirement	Where to check in cPanel
PHP **8.3** (8.1 minimum)	*MultiPHP Manager*
PHP extensions: `pdo_mysql`, `zip`, `curl`, `openssl`, `fileinfo`, `dom`	*Select PHP Version* or provider
MariaDB 10.6+ or MySQL 8.0+	*MySQL Databases*
Apache `mod_rewrite` and `.htaccess`	Provider (enabled on standard cPanel)
SSL certificate for the subdomain	*SSL/TLS Status* (AutoSSL)
Disk space: about 200 MB, plus data	*Disk Usage*
Files (inodes): about 16,000	Provider

Ask your provider one question

> *What is the maximum number of simultaneous connections per database user (`max_user_connections`) on our plan?*

A value of **20 or more** is suitable. Each synchronization uses one connection for a fraction of a second.

What you need

- The package `csweb-<version>-cpanel.zip` and its checksum file `.sha256`.
- A subdomain name, for example `csweb.your-domain.org`. This guide uses this example.
- A strong administrator password (12 characters or more).

Installation

Step 1. Create the subdomain

1. In cPanel, open **Domains** (or **Subdomains**) and create `csweb.your-domain.org`.
2. Set its **document root** to `/home/<cpanel-user>/csweb`.

   Do not use a folder inside `public_html`. The package is extracted into `/home/<cpanel-user>/csweb` in Step 4.

Step 2. Select the PHP version and settings

1. Open **MultiPHP Manager**, select the subdomain and choose **PHP 8.3**.
2. Open **MultiPHP INI Editor**, select the subdomain and set:

Setting	Value
`memory_limit`	`256M`
`upload_max_filesize`	`64M`
`post_max_size`	`64M`
`max_execution_time`	`300`

Step 3. Create the database

1. Open **MySQL Databases**.
2. Create a database, for example `csweb`. cPanel adds your account prefix: note the full name, for example `myaccount_csweb`.
3. Create a user, for example `myaccount_cswebuser`, with a strong password.
4. Under **Add User To Database**, select the user and the database, then grant **ALL PRIVILEGES**.

Step 4. Upload and extract the package

1. Open **File Manager** and go to your **home folder** (`/home/<cpanel-user>`), not `public_html`.
2. Click **Upload** and select `csweb-<version>-cpanel.zip`.
3. Back in File Manager, right-click the zip and choose **Extract**. This creates the folder `csweb`.
4. Click **Settings** (top right) and tick **Show Hidden Files (dotfiles)**. Open `csweb` and check that the file `.htaccess` is present. Without it, CSEntry cannot authenticate.
5. Delete the zip file.

> **Optional: check the download.** Compare the SHA-256 of the zip with the `.sha256` file before uploading (on Windows: `certutil -hashfile file.zip SHA256`).

Folder permissions are already correct: on cPanel, PHP runs as your account, so **do not set 777 permissions**.

Step 5. Enable HTTPS

1. Open **SSL/TLS Status**, select `csweb.your-domain.org` and click **Run AutoSSL**.
2. Wait until `https://csweb.your-domain.org` opens without a certificate warning. A page from the CSWeb setup is expected at this stage.

Complete this step **before** Step 6: the address entered in the setup wizard is the one written into the CSEntry applications.

Step 6. Run the setup wizard

1. Open `https://csweb.your-domain.org/setup/`. All requirements must show a green check. If one is red, see *Troubleshooting*.
2. Click **Next** and fill in the form:

Field	Value
Database name	Full name from Step 3, for example `myaccount_csweb`
Database host	`localhost`
Database user / password	From Step 3
Admin password	Your strong administrator password
Timezone	Your local timezone, for example `Asia/Baghdad`
Max execution time	`300`
Files directory	`/home/<cpanel-user>/csweb/files` (pre-filled)
CSWeb API URL	`https://csweb.your-domain.org/api/` **(with the final slash)**

3. Click **Next**. The wizard creates the database and shows a completion page.

After installation, the wizard **locks itself** (`/setup/` returns *403 Forbidden*). This is expected and protects your installation.

Step 7. Validate the installation

Complete this checklist before using the server for fieldwork.

#	Test	Expected result
1	Log in at `https://csweb.your-domain.org` as `admin`	The dashboard opens, with styled tables and menus
2	**Users**: create an interviewer account	The account appears in the list
3	From **CSPro Designer**, deploy an application to `https://csweb.your-domain.org/api`	The application appears under **Apps**
4	On a device, in **CSEntry**, add the application from the server, log in as the interviewer	The application installs
5	Enter 2 or 3 test cases and **synchronize**, preferably over mobile data	Sync completes; cases appear in **Sync Report**
6	Synchronize a second device	It downloads the shared files and data as designed
7	Download the data with **CSPro Data Viewer** (download from CSWeb, using the server address above)	All test cases are present

Then delete the test data from the dashboard's list of dictionaries (*Delete data*), so that fieldwork starts from an empty database.

Operation

Backups

Use the **cPanel backups**. CSWeb's own *Backup now* button needs functions that shared hosting disables, and shows an error.

- **Backup** (or **JetBackup**) in cPanel saves the database and files daily on most plans. Check the frequency and retention with your provider.
- During fieldwork, download a copy at least weekly: **Backup > Download a MySQL Database Backup** (your CSWeb database) and **Download a Home Directory Backup**.
- Data is also kept on the devices until it is synchronized. Do not delete CSEntry data on devices until it has been downloaded and checked.

Accounts and security

- Use one account per interviewer. The server limits repeated wrong passwords: after **5 failed attempts in one minute**, an account is locked for one minute.
- Keep the `admin` password private. Create supervisor accounts for daily monitoring.
- Always use the `https://` address in CSPro Designer and CSEntry.

Capacity

Situation	Behaviour
Up to 200 devices synchronizing within the same minute	Tested with no error
More than 400 devices within the same minute	A few syncs may fail and must be retried
Devices sharing one network (office Wi-Fi, mobile operator NAT)	Supported (each interviewer has their own account)
Large syncs (5,000 cases at once)	Tested: about 2 seconds

For a census or a survey with several hundred interviewers synchronizing at the same time, a dedicated server is recommended.

Where to find logs

`/home/<cpanel-user>/csweb/var/logs/` (`api-<date>.log` for synchronization, `ui.log` for the web interface). These files are not accessible from the web.

Troubleshooting

Symptom	Cause and solution
A requirement is red in the setup wizard	PHP version or extension: see Step 2 and the hosting checklist. For "directories are writeable", re-extract the zip with the File Manager.
HTTP 500 when logging in to the dashboard right after setup	The API URL lacks its final slash. In File Manager, edit `csweb/src/config.php` and make the `API_URL` line end with `/api/'`.
Pages display without style; buttons and tables do not work	Incomplete upload. Delete the `csweb` folder and extract the full zip again.
CSEntry or Designer: *Unauthorized* (401) with correct passwords	The hidden file `csweb/.htaccess` is missing. Re-extract the zip (Step 4).
Too many requests (429)	Repeated wrong passwords: wait one minute.
Some syncs fail at peak times; the log shows `max_user_connections`	The plan's connection limit is reached. Ask the provider to raise it, or schedule teams to sync at different times.
Large syncs fail or time out	Check Step 2 settings (`max_execution_time`, upload limits). If the provider's firewall (ModSecurity) blocks requests, ask them to allow `/api/` on the CSWeb subdomain.
Backup now shows an error	Expected on shared hosting: use the cPanel backups.
Settings, Backup or Logs page shows an error	These pages are prepared automatically on the first visit after installation: reload the page. If the error persists, see `var/logs/ui.log`.
`/setup/` returns *403 Forbidden*	Normal after installation.

Updating CSWeb

1. Make a cPanel backup (database and home directory).
2. Extract the new package into a new folder, for example `csweb-new`.
3. Copy into it, from the current installation: the file `src/config.php` (written by the setup wizard) and the folders `files/` and `files_csweb/`.
4. Swap the folders (rename `csweb` to `csweb-old`, then `csweb-new` to `csweb`), and empty the folder `csweb/var/cache`.
5. Run the checklist of Step 7. Keep `csweb-old` until the update is validated.

Support

CSWeb Community: https://github.com/BOUNADRAME/csweb-community. Report issues under *Issues* with the content of `BUILD-INFO.txt` and the relevant lines of `var/logs/`.

Installation assistance and dedicated server set-up are available from StatPro Consulting.